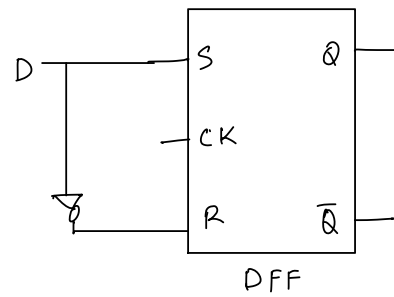
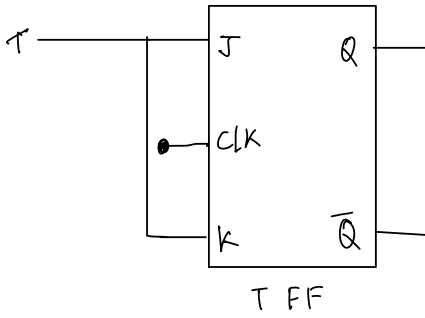


- Design a D FF using SR FF.
- Design a T FF using JK FF.
- Design a D FF using JK FF.
- Design a SR FF.

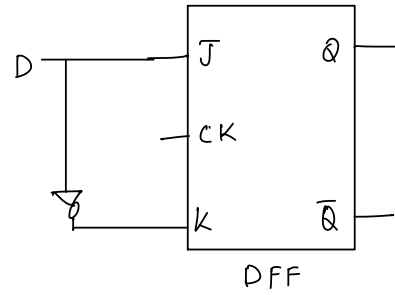
①



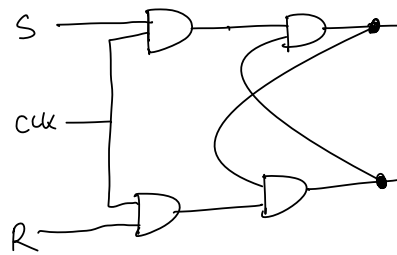
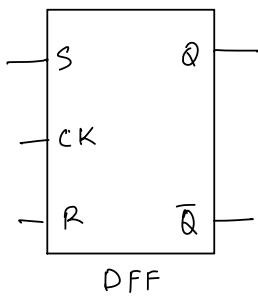
②



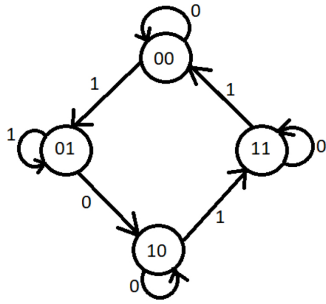
③



④



Given the state diagram as follows, get the sequential circuit using SR flipflop.



Ans:

| A | B | x | A <sup>+</sup> | B <sup>+</sup> | S <sub>A</sub> | R <sub>A</sub> | S <sub>B</sub> | R <sub>B</sub> |
|---|---|---|----------------|----------------|----------------|----------------|----------------|----------------|
| 0 | 0 | 0 | 0              | 0              | 0              | X              | 0              | X              |
| 0 | 0 | 1 | 0              | 1              | 0              | X              | 1              | 0              |
| 0 | 1 | 0 | 1              | 0              | 1              | 0              | 0              | 1              |
| 0 | 1 | 1 | 0              | 1              | 0              | X              | X              | 0              |
| 1 | 0 | 0 | 1              | 0              | X              | 0              | 0              | X              |
| 1 | 0 | 1 | 1              | 1              | X              | 0              | 1              | 0              |
| 1 | 1 | 0 | 1              | 1              | X              | 0              | X              | 0              |
| 1 | 1 | 1 | 0              | 0              | 0              | 1              | 0              | 1              |

for S<sub>A</sub>,

| A  | Bx | B'x' | B'x | Bx | Bx' |
|----|----|------|-----|----|-----|
| A' |    |      |     |    | 1   |
| A  | X  |      | X   |    | X   |

S<sub>A</sub> = Bx'

for S<sub>B</sub>,

| A  | Bx | B'x' | B'x | Bx | Bx' |
|----|----|------|-----|----|-----|
| A' |    |      | 1   | X  |     |
| A  |    |      | 1   |    | X   |

S<sub>B</sub> = B'x

for R<sub>A</sub>,

| A  | Bx | B'x' | B'x | Bx | Bx' |
|----|----|------|-----|----|-----|
| A' | X  |      | X   |    |     |
| A  |    |      |     | 1  |     |

R<sub>A</sub> = Bx

for R<sub>B</sub>,

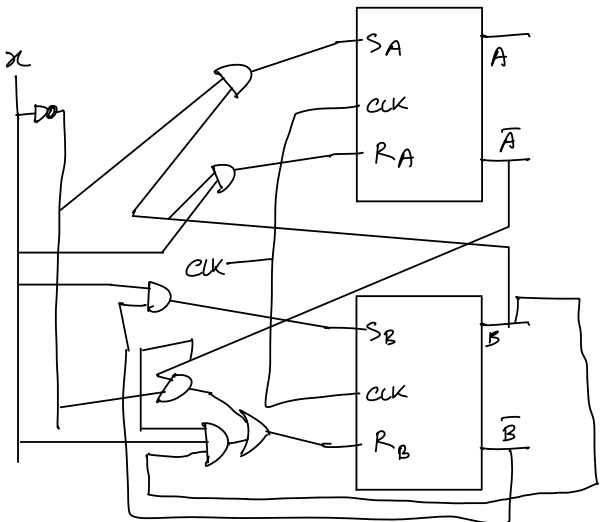
| A  | Bx | B'x' | B'x | Bx | Bx' |
|----|----|------|-----|----|-----|
| A' | X  |      |     |    | 1   |
| A  | X  |      |     | 1  |     |

R<sub>B</sub> = A'x' + ABx

R<sub>A</sub> = Bx

S<sub>B</sub> = B'x

S<sub>A</sub> = Bx'

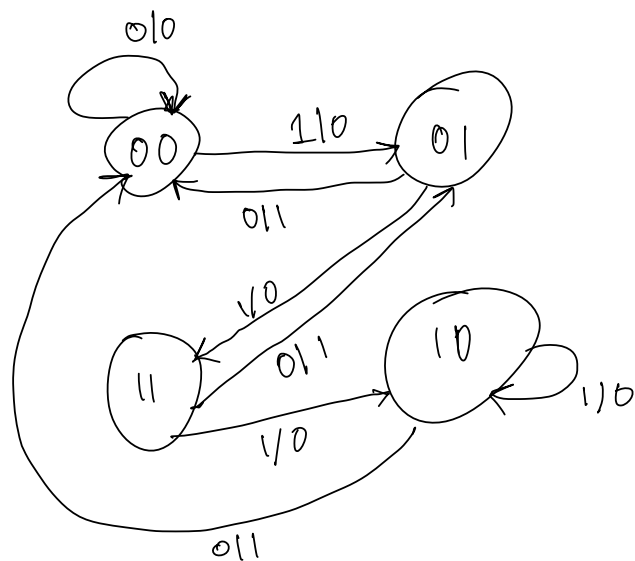


- 
- The diagram shows a logic circuit with the following components and connections:
- Inputs:**  $x$ ,  $y$ , and a common clock signal  $CLK$ .
  - Logic Gates:**
    - Two 2-to-1 multiplexers (labeled  $A \times 2$  and  $B \times 2$ ).
    - Two AND gates.
    - One OR gate.
    - One NOT gate.
  - Flip-Flops:** Two D flip-flops, one for output  $A$  and one for output  $B$ .
  - Connections:**
    - The clock signal  $CLK$  is connected to the clock inputs of both flip-flops.
    - The output  $A$  of the first flip-flop is connected to the top input of the  $A \times 2$  multiplexer and the top input of the OR gate.
    - The output  $B$  of the second flip-flop is connected to the bottom input of the  $A \times 2$  multiplexer, the top input of the  $B \times 2$  multiplexer, and the top input of the AND gate that produces  $y$ .
    - The input  $x$  is connected to the top input of the  $B \times 2$  multiplexer and the top input of the NOT gate.
    - The output  $y$  is connected to the bottom input of the  $B \times 2$  multiplexer and the bottom input of the AND gate that produces  $y$ .
    - The output of the  $A \times 2$  multiplexer is connected to the D input of the first flip-flop.
    - The output of the  $B \times 2$  multiplexer is connected to the D input of the second flip-flop.
    - The output of the OR gate is connected to the  $\bar{Q}$  output of the first flip-flop.
    - The output of the NOT gate is connected to the bottom input of the AND gate that produces  $y$ .

$$D_3 = \overline{A}x$$

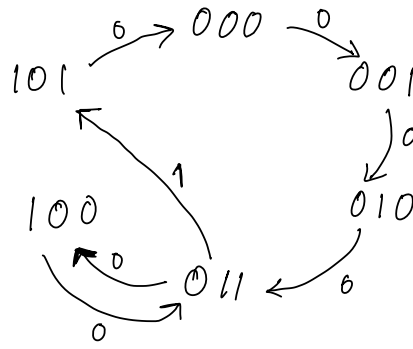
$$y = (A+B) \cdot \bar{x}$$

| A | B | x | D <sub>A</sub> | D <sub>B</sub> | y | A <sup>+</sup> | B <sup>+</sup> |
|---|---|---|----------------|----------------|---|----------------|----------------|
| 0 | 0 | 0 | 0              | 0              | 0 | 0              | 0              |
| 0 | 0 | 1 | 0              | 1              | 0 | 0              | 1              |
| 0 | 1 | 0 | 0              | 0              | 1 | 0              | 0              |
| 0 | 1 | 1 | 1              | 1              | 0 | 1              | 1              |
| 1 | 0 | 0 | 0              | 0              | 1 | 0              | 0              |
| 1 | 0 | 1 | 1              | 0              | 0 | 1              | 0              |
| 1 | 1 | 0 | 0              | 0              | 1 | 0              | 1              |
| 1 | 1 | 1 | 1              | 0              | 0 | 1              | 0              |



$CSE110 \rightarrow CSE111 \rightarrow CSE220 \rightarrow CSE221 \rightarrow CSE331 \rightarrow CSE221 \rightarrow CSE321 \rightarrow CSE110$   
 1                      2                      3                      4                      5                      4                      6                      1

$CSE110 \rightarrow 000$   
 $CSE111 \rightarrow 001$   
 $CSE220 \rightarrow 010$   
 $CSE221 \rightarrow 011$   
 $CSE331 \rightarrow 100$   
 $CSE321 \rightarrow 101$



|   |   |   |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

| $x$ | A | B | C | $A^+$ | $B^+$ | $C^+$ | $T_A$ | $T_B$ | $T_C$ |    |
|-----|---|---|---|-------|-------|-------|-------|-------|-------|----|
| 0   | 0 | 0 | 0 | 0     | 0     | 1     | 0     | 0     | 1     | 0  |
| 0   | 0 | 0 | 1 | 0     | 1     | 0     | 0     | 1     | 1     | 1  |
| 0   | 0 | 1 | 0 | 0     | 1     | 1     | 0     | 0     | 1     | 2  |
| 0   | 0 | 1 | 1 | 1     | 0     | 0     | 1     | 1     | 1     | 3  |
| 0   | 1 | 0 | 0 | 0     | 1     | 1     | 1     | 1     | 1     | 4  |
| 0   | 1 | 0 | 1 | 0     | 0     | 0     | 1     | 0     | 1     | 5  |
| 0   | 1 | 1 | 0 | X     | X     | X     | X     | X     | X     | 6  |
| 0   | 1 | 1 | 1 | X     | X     | X     | X     | X     | X     | 7  |
| 1   | 0 | 0 | 0 | 0     | 0     | 1     | 0     | 0     | 1     | 8  |
| 1   | 0 | 0 | 1 | 0     | 1     | 0     | 0     | 1     | 1     | 9  |
| 1   | 0 | 1 | 0 | 0     | 1     | 1     | 0     | 0     | 1     | 10 |
| 1   | 0 | 1 | 1 | 1     | 0     | 1     | 1     | 1     | 0     | 11 |
| 1   | 1 | 0 | 0 | 0     | 1     | 1     | 1     | 1     | 1     | 12 |
| 1   | 1 | 0 | 1 | 0     | 0     | 0     | 1     | 0     | 1     | 13 |
| 1   | 1 | 1 | 0 | X     | X     | X     | X     | X     | X     | 14 |
| 1   | 1 | 1 | 1 | X     | X     | X     | X     | X     | X     | 15 |

For  $T_A$ ,

| $xA \backslash BC$ | $B'C'$ | $B'C$ | $BC$ | $BC'$ |
|--------------------|--------|-------|------|-------|
| $x'A'$             | 0      | 1     | 3    | 2     |
| $x'A$              | 4      | 5     | 7    | 6     |
| $xA$               | 12     | 13    | 15   | 14    |
| $x'A'$             | 8      | 9     | 11   | 10    |

$$T_A = A + BC$$

For  $T_B$ ,

| $xA \backslash BC$ | $B'C'$ | $B'C$ | $BC$ | $BC'$ |
|--------------------|--------|-------|------|-------|
| $x'A'$             | 0      | 1     | 3    | 2     |
| $x'A$              | 4      | 5     | 7    | 6     |
| $xA$               | 12     | 13    | 15   | 14    |
| $x'A'$             | 8      | 9     | 11   | 10    |

$$T_B = AC' + A'C$$

$$\Rightarrow T_B = A \oplus C$$

For  $T_C$ ,

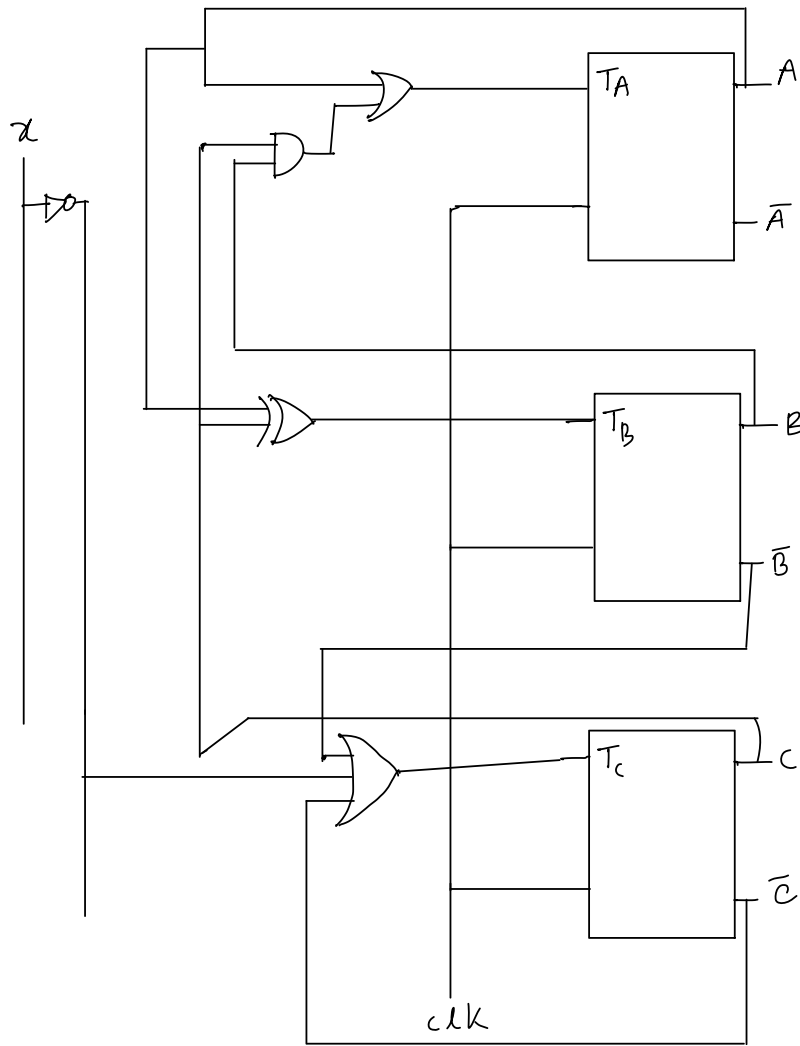
| $xA \backslash BC$ | $B'C'$ | $B'C$ | $BC$ | $BC'$ |
|--------------------|--------|-------|------|-------|
| $x'A'$             | 0      | 1     | 3    | 2     |
| $x'A$              | 4      | 5     | 7    | 6     |
| $xA$               | 12     | 13    | 15   | 14    |
| $x'A'$             | 8      | 9     | 11   | 10    |

$$T_C = B' + x' + C'$$

$$T_C = B' + x' + C'$$

$$T_A = A + BC$$

$$\Rightarrow T_B = A \oplus C$$

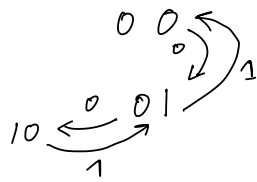


- Implement the following counter using JK FF: Green->Yellow->Red->Yellow->Green

Green  $\rightarrow 00$

Yellow  $\rightarrow 01$

Red  $\rightarrow 10$



For  $J_A$ ,

| $x \backslash AB$ | $A'B'$ | $A'B$ | $AB$ | $AB'$ |
|-------------------|--------|-------|------|-------|
| $x'$              |        | 1     | X    | X     |
| $x$               |        |       | X    | X     |

$$J_A = x'B$$

|   | $x$ | $A$ | $B$ | $A^+$ | $B^+$ | $J_A$ | $K_A$ | $J_B$ | $K_B$ |
|---|-----|-----|-----|-------|-------|-------|-------|-------|-------|
| 0 | 0   | 0   | 0   | 0     | 1     | 0     | X     | 1     | X     |
| 1 | 0   | 0   | 1   | 1     | 0     | 1     | X     | X     | 1     |
| 2 | 0   | 1   | 0   | 0     | 1     | X     | 1     | 1     | X     |
| 3 | 0   | 1   | 1   | X     | X     | X     | X     | X     | X     |
| 4 | 1   | 0   | 0   | 0     | 1     | 0     | X     | 1     | X     |
| 5 | 1   | 0   | 1   | 0     | 0     | 0     | X     | X     | 1     |
| 6 | 1   | 1   | 0   | 0     | 1     | X     | 1     | 1     | X     |
| 7 | 1   | 1   | 1   | X     | X     | X     | X     | X     | X     |

For  $K_A$ ,

| $\chi$ \ $AB$ | $A'B'$         | $A'B$          | $AB$           | $AB'$          |
|---------------|----------------|----------------|----------------|----------------|
| $\chi'$       | X <sub>0</sub> | X <sub>1</sub> | X <sub>3</sub> | 1 <sub>2</sub> |
| $\chi$        | X <sub>4</sub> | X <sub>5</sub> | X <sub>7</sub> | 1 <sub>6</sub> |

$$K_A = 1$$

For  $J_B$ ,

| $\chi$ \ $AB$ | $A'B'$         | $A'B$          | $AB$           | $AB'$          |
|---------------|----------------|----------------|----------------|----------------|
| $\chi'$       | / <sub>0</sub> | X <sub>1</sub> | X <sub>3</sub> | / <sub>2</sub> |
| $\chi$        | / <sub>4</sub> | X <sub>5</sub> | X <sub>7</sub> | 1 <sub>6</sub> |

$$J_B = 1$$

For  $K_B$ ,

| $\chi$ \ $AB$ | $A'B'$         | $A'B$          | $AB$           | $AB'$          |
|---------------|----------------|----------------|----------------|----------------|
| $\chi'$       | X <sub>0</sub> | 1 <sub>1</sub> | X <sub>3</sub> | X <sub>2</sub> |
| $\chi$        | X <sub>4</sub> | 1 <sub>5</sub> | X <sub>7</sub> | X <sub>6</sub> |

$$K_B = 1$$

